

CURSORS

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```
Database changed
mysql> CREATE TABLE Student (
  ->     SID INT PRIMARY KEY,
  ->     SName VARCHAR(50),
  ->     Marks INT
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Student VALUES
  -> (101, 'Rahul', 85),
  -> (102, 'Priya', 78),
  -> (103, 'Arun', 92),
  -> (104, 'Sneha', 88),
  -> (105, 'Kiran', 75);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Student;
+-----+-----+-----+
| SID | SName | Marks |
+-----+-----+-----+
| 101 | Rahul | 85    |
| 102 | Priya | 78    |
| 103 | Arun  | 92    |
| 104 | Sneha | 88    |
| 105 | Kiran | 75    |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

SIMPLE CURSOR

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE SimpleCursor()
  -> BEGIN
  ->     DECLARE id INT;
  ->     DECLARE name VARCHAR(50);
  ->
  ->     DECLARE student_cursor CURSOR FOR
  ->         SELECT SID, SName FROM Student;
  ->
  ->     OPEN student_cursor;
  ->
  ->     FETCH student_cursor INTO id, name;
  ->
  ->     SELECT id AS StudentID, name AS StudentName;
  ->
  ->     CLOSE student_cursor;
  -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL SimpleCursor();
+-----+-----+
| StudentID | StudentName |
+-----+-----+
| 101 | Rahul |
+-----+-----+
1 row in set (0.01 sec)
```

CURSOR WITH LOOP

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE DisplayStudents()
-> BEGIN
->     DECLARE id INT;
->     DECLARE name VARCHAR(50);
->     DECLARE done BOOLEAN DEFAULT FALSE;
->
->     DECLARE student_cursor CURSOR FOR
->         SELECT SID, SName FROM Student;
->
->     DECLARE CONTINUE HANDLER FOR NOT FOUND
->         SET done = TRUE;
->
->     OPEN student_cursor;
->
->     read_loop: LOOP
->         FETCH student_cursor INTO id, name;
->
->         IF done THEN
->             LEAVE read_loop;
->         END IF;
->
->         SELECT id AS StudentID, name AS StudentName;
->
->     END LOOP read_loop;
->
->     CLOSE student_cursor;
->
-> END //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CALL DisplayStudents();
+-----+-----+
| StudentID | StudentName |
+-----+-----+
|         101 | Rahul       |
+-----+-----+
1 row in set (0.00 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|         102 | Priya      |
+-----+-----+
1 row in set (0.01 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|         103 | Arun       |
+-----+-----+
1 row in set (0.01 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|         104 | Sneha      |
+-----+-----+
1 row in set (0.01 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|         105 | Kiran      |
+-----+-----+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```

CURSOR WITH WHILE

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE CursorWhile()
-> BEGIN
->     DECLARE id INT;
->     DECLARE name VARCHAR(50);
->     DECLARE done BOOLEAN DEFAULT FALSE;
->
->     DECLARE student_cursor CURSOR FOR
->         SELECT SID, SName FROM Student;
->
->     DECLARE CONTINUE HANDLER FOR NOT FOUND
->         SET done = TRUE;
->
->     OPEN student_cursor;
->
->     FETCH student_cursor INTO id, name;
->
->     WHILE done = FALSE DO
->         SELECT id AS StudentID, name AS StudentName;
->
->         FETCH student_cursor INTO id, name;
->
->     END WHILE;
->
->     CLOSE student_cursor;
->
-> END //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql> CALL CursorWhile();
+-----+-----+
| StudentID | StudentName |
+-----+-----+
|      101 | Rahul      |
+-----+-----+
1 row in set (0.00 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|      102 | Priya      |
+-----+-----+
1 row in set (0.00 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|      103 | Arun       |
+-----+-----+
1 row in set (0.01 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|      104 | Sneha      |
+-----+-----+
1 row in set (0.01 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|      105 | Kiran      |
+-----+-----+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```

CURSOR WITH REPEAT

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE CursorRepeat()
-> BEGIN
->     DECLARE id INT;
->     DECLARE name VARCHAR(50);
->     DECLARE done BOOLEAN DEFAULT FALSE;
->
->     DECLARE student_cursor CURSOR FOR
->         SELECT SID, SName FROM Student;
->
->     DECLARE CONTINUE HANDLER FOR NOT FOUND
->         SET done = TRUE;
->
->     OPEN student_cursor;
->
->     REPEAT
->
->         FETCH student_cursor INTO id, name;
->
->         IF NOT done THEN
->             SELECT id AS StudentID, name AS StudentName;
->         END IF;
->
->     UNTIL done
->     END REPEAT;
->
->     CLOSE student_cursor;
->
-> END //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql> CALL CursorRepeat();
+-----+-----+
| StudentID | StudentName |
+-----+-----+
|      101 | Rahul      |
+-----+-----+
1 row in set (0.00 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|      102 | Priya      |
+-----+-----+
1 row in set (0.00 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|      103 | Arun       |
+-----+-----+
1 row in set (0.01 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|      104 | Sneha      |
+-----+-----+
1 row in set (0.01 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|      105 | Kiran      |
+-----+-----+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```

CURSOR WITH UPDATE

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE AddBonus()  
-> BEGIN  
->   DECLARE id INT;  
->   DECLARE done BOOLEAN DEFAULT FALSE;  
->  
->   DECLARE student_cursor CURSOR FOR  
->     SELECT SID FROM Student;  
->  
->   DECLARE CONTINUE HANDLER FOR NOT FOUND  
->     SET done = TRUE;  
->  
->   OPEN student_cursor;  
->  
->   read_loop: LOOP  
->  
->     FETCH student_cursor INTO id;  
->  
->     IF done THEN  
->       LEAVE read_loop;  
->     END IF;  
->  
->     UPDATE Student  
->       SET Marks = Marks + 5  
->       WHERE SID = id;  
->  
->   END LOOP;  
->  
->   CLOSE student_cursor;  
->  
-> END //
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql> CALL AddBonus();
```

```
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> SELECT * FROM Student;
```

SID	SName	Marks
101	Rahul	90
102	Priya	83
103	Arun	97
104	Sneha	93
105	Kiran	80

```
5 rows in set (0.00 sec)
```

```
mysql> |
```